

# Rhythm of Opinion: Interpretable Hawkes-Graph Networks for Hierarchical Opinion Propagation

Yulong Li\*

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Yulong.Li19@student.xjtlu.edu.cn

Zhixiang Lu\*

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Zhixiang.Lu22@student.xjtlu.edu.cn

Peixin Guo

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Peixin.Guo22@student.xjtlu.edu.cn

Simin Lai

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Simin.Lai23@student.xjtlu.edu.cn

Yuxuan Zhang

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Yuxuan.Zhang2104@student.xjtlu.edu.cn

Haochen Xue

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Haochen.Xue20@student.xjtlu.edu.cn

Xiwei Liu

MBZUAI  
Abu Dhabi, United Arab Emirates  
Xiwei.Liu@mbzuai.ac.ae

Yichen Li

Huazhong University of Science and  
Technology  
Wuhan, China  
ycli0204@hust.edu.cn

Zhaodong Wu

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Zhaodong.Wu@student.xjtlu.edu.cn

Feilong Tang

Monash University  
Melbourne, Australia  
Feilong.Tang@monash.edu

Mian Zhou

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Mian.Zhou@xjtlu.edu.cn

Chong Li

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Chong.Li@xjtlu.edu.cn

Imran Razzak†

MBZUAI  
Abu Dhabi, United Arab Emirates  
Imran.Razzak@mbzuai.ac.ae

Qingxia Li†

Fisk University  
Tennessee, United States of America  
qli@fisk.edu

Jionglong Su†

Xi'an Jiaotong-Liverpool University  
Suzhou, Jiang Su, China  
Jionglong.Su@xjtlu.edu.cn

## Abstract

Opinion propagation research primarily focuses on phenomenon prediction rather than mechanism understanding, lacking interpretable frameworks to reveal underlying propagation dynamics. This limitation stems from two sources: existing methods employ end-to-end paradigms where parameters lack physical meanings, while available datasets suffer from incomplete hierarchical structures, coarse sentiment annotations, and limited domain coverage. To address these limitations, we introduce VISTA, a multi-dimensional opinion propagation dataset providing complete hierarchical structures, fine-grained emotional annotations, and cross-domain coverage. Based on this dataset, we propose an interpretable modeling framework integrating high-dimensional Hawkes processes with graph neural networks, enabling parametric expression of propagation mechanisms through event space constructed from emotional and reply level combinations. Through interpretable parameter analysis, we reveal three mechanistic patterns: differential emotional propagation strength, asymmetric hierarchical excitation, and temporal memory effects. Our framework establishes quantitative foundations for understanding opinion propagation dynamics, achieving best performance in sentiment prediction and structural consistency tasks while providing the first benchmark for multi-dimensional propagation mechanism analysis.

## CCS Concepts

• **Applied computing** → *Sociology*.

## Keywords

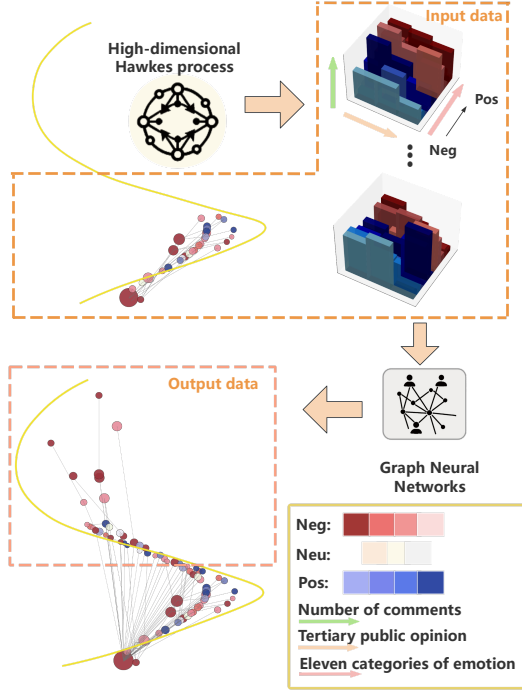
Opinion propagation; Graph; Hawkes Processes

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## 1 Introduction

Opinion propagation represents a key scientific problem in understanding collective behavior emergence through dynamic mechanism analysis. However, current research focuses on phenomenon prediction rather than mechanism understanding [9, 30, 39, 56], lacking methods to reveal multi-dimensional, dynamic interactions from observational data [4, 32]. This limitation restricts opinion science development, as effective intervention, cross-domain transfer, and transparent explanations require deep mechanistic insights [19, 34]. Understanding opinion propagation mechanisms faces three challenges: nonlinear dynamics in multi-level networks [10, 11, 27],



**Figure 1: Opinion Dynamics Modeling integrates high-dimensional Hawkes processes with graph neural network-based sentiment analysis to capture public opinion dynamics.**

complex memory effects in opinion evolution [14], and temporal decay patterns in multi-dimensional propagation [42, 48]. Addressing these challenges requires high-quality data supporting multi-dimensional analysis. However, existing datasets' limitations in structural completeness, annotation granularity, and domain diversity provide insufficient support for quantitative modeling and parameter interpretation, hindering the transition from statistical fitting to mechanism explanation. Opinion propagation represents a key scientific problem in understanding collective behavior emergence through dynamic mechanism analysis. However, current research focuses on phenomenon prediction rather than mechanism understanding [9, 30, 39, 56], lacking methods to reveal multi-dimensional, dynamic interactions from observational data [4, 32]. This limitation restricts opinion science development, as effective intervention, cross-domain transfer, and transparent explanations require deep mechanistic insights [19, 34]. Understanding opinion propagation mechanisms faces three challenges: nonlinear dynamics in multi-level networks [10, 11, 27], complex memory effects in opinion evolution [14, 29, 31], and temporal decay patterns in multi-dimensional propagation [42, 48]. Addressing these challenges requires high-quality data supporting multi-dimensional analysis. However, existing datasets are inadequate in structural completeness, annotation granularity, and domain diversity, hindering quantitative modeling and the transition from statistical fitting to mechanistic explanation.

Existing datasets limit quantitative analysis of multi-dimensional propagation mechanisms. *i) Structural completeness:* Available datasets lack complete multi-level reply hierarchies and lifecycle records

**Table 1: Multidimensional comparison of datasets**

| Dataset                      | Md | Cd | Lt | Cpoe | El |
|------------------------------|----|----|----|------|----|
| Twitter Dataset [45]         | ✓  | ✓  | ✗  | ✗    | ✓* |
| Reddit Dataset [6]           | ✓  | ✓  | ✗  | ✗    | ✗  |
| YouTube Dataset [1]          | ✓  | ✓  | ✗  | ✗    | ✗  |
| FakeNewsNet Dataset [43]     | ✗  | ✗  | ✓  | ✗    | ✓  |
| Political Event Dataset [16] | ✗  | ✗  | ✗  | ✗    | ✗  |
| COVID-19 Dataset [15]        | ✗  | ✗  | ✓  | ✗    | ✗  |
| <b>VISTA (Ours)</b>          | ✓  | ✓  | ✓  | ✓    | ✓  |

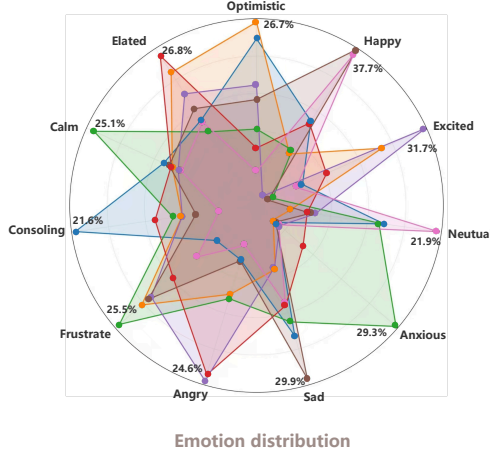
Md: Multilevel data. Cd: Content diversity. Lt: Length of time. Cpoe: Complete public opinion event. El: Emotional label. \*:Partial annotation

from topic emergence to decay, failing to support parameter estimation of nonlinear dynamics and temporal memory effects. *ii) Annotation granularity:* Current datasets employ five-element or six-element sentiment classifications, unable to provide high-dimensional emotional state spaces required for modeling complex memory effects. *iii) Domain diversity:* Single-topic compositions cannot provide multi-domain empirical samples required for validating cross-domain propagation hypotheses. These deficiencies prevent establishing parameter-phenomenon mappings, limiting the transition from statistical fitting to mechanism explanation.

To address these limitations, we propose the VISTA dataset, an opinion propagation dataset simultaneously possessing five dimensions: multi-level data, content diversity, temporal length, complete opinion events, and emotional annotations. As shown in Table 1, existing datasets suffer from dimensional deficiencies, unable to support systematic modeling of multi-dimensional propagation mechanisms. VISTA contains 159 cross-domain topics, 47,207 root posts, 327,015 second-level comments, and 29,578 third-level comments, forming a complete three-level hierarchical network supporting parametric modeling of multi-level dynamics and temporal memory effects. The dataset employs an 11-category emotional annotation system with Cohen's Kappa of 0.85, providing high-dimensional feature space for complex state transition analysis. VISTA records complete topic lifecycles from emergence to decay, providing temporal support for long-term memory effect modeling. This multidimensional completeness makes VISTA a benchmark for quantitative analysis of multidimensional propagation mechanisms.

Current modeling methods have limitations in mechanism interpretation capabilities. *i) Parameter interpretability:* Methods such as GCN [26], LSTM [52], and Transformer [24] adopt end-to-end prediction paradigms, where parameters lack clear mechanism meanings and cannot be interpreted as physically meaningful propagation parameters such as excitation intensity and decay coefficients. *ii) Multi-dimensional interaction modeling:* Methods such as GAT [40] and GraphSAINT [12] treat emotional propagation and hierarchical structures as independent features, lacking frameworks for modeling interaction effects within unified parameter spaces. *iii) Temporal dynamics modeling:* Temporal models such as ARIMA [7] and RNN [17] cannot simultaneously capture coupling mechanisms of short-term excitation and long-term memory decay. These limitations prevent direct extraction of mechanistic insights from model parameters in practice, thereby hindering the transition from prediction to rigorous mechanism explanation.

To address these limitations, We propose an interpretable coupling of high-dimensional Hawkes processes and graph neural networks. This framework constructs high-dimensional event space



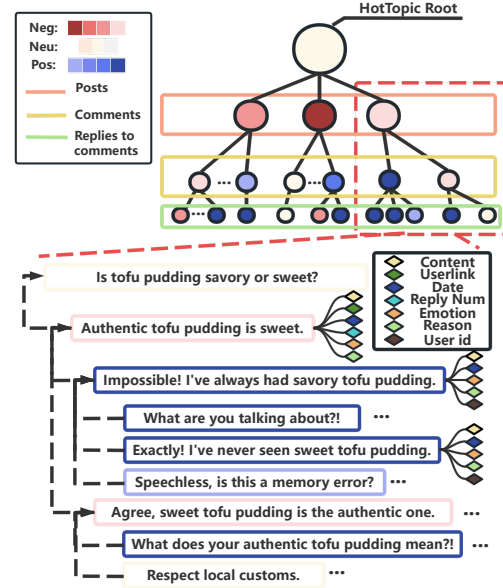
**Figure 2: Distribution of 11 emotional categories in the VISTA**

through Cartesian product of emotional and reply levels, enabling each parameter to correspond to clear mechanism meanings: excitation parameters quantify propagation strength between emotion-level combinations, while decay parameters characterize temporal properties of memory effects. The framework models interaction effects of emotional propagation and hierarchical diffusion within unified parameter space, avoiding independent feature processing limitations. Based on Hawkes process predictions, graph neural networks model structural propagation paths, achieving unified temporal-spatial modeling. Through interpretable parameter analysis on VISTA, we discovered mechanistic patterns: differential emotional propagation strength, asymmetric hierarchical excitation, and temporal memory effects, providing quantitative foundations for understanding opinion propagation dynamics.

Our contributions are as follows: 1) We propose an interpretable opinion propagation modeling framework integrating high-dimensional Hawkes processes with graph neural networks, achieving parametric expression of propagation mechanisms. 2) We introduce the VISTA dataset, a large-scale opinion propagation dataset simultaneously possessing five dimensions, filling the data gap in multi-dimensional propagation mechanism modeling. 3) We establish a rigorous quantitative analysis paradigm that systematically reveals differential emotional propagation strength, asymmetric hierarchical excitation, and temporal memory effects.

## 2 Related Work

Opinion propagation has undergone substantial development since the mid-20th century, initially grounded in classical sociological theories, including Katz and Lazarsfeld’s Two-Step Flow Theory [23], which emphasizes the role of opinion leaders in information dissemination, and Noelle-Neumann’s Spiral of Silence Theory [36], which accounts for the silence of minority opinions. With the rise of the internet, time series models such as AutoRegressive Integrated Moving Average (ARIMA) have been employed to forecast popularity trajectories in opinion events, although they struggle with pronounced nonlinear relationships [7, 17]. The Susceptible-Infected-Recovered (SIR) model [51] is adapted for opinion propagation but assumes static networks, limiting its applicability in dynamic contexts [2, 51]. Complex network theories, such as Barabási



**Figure 3: Illustrative Diagram of VISTA Datasets.**

and Albert’s scale-free network [5] as well as Watts and Strogatz’s small-world theory [49], provide further insights into network structures. However, methods such as Graph Convolutional Networks (GCNs) [26] also assume static structures and cannot fully capture evolving dynamics [26, 52]. Hierarchical models [22, 53] address stepwise information spread, but lack temporal dynamics [3, 35].

From 2015 to 2020, deep learning algorithm such as LSTM and RNN models [52] achieve a breakthrough in modeling opinion propagation, capturing long-term dependencies but neglecting opinion structure. GNNs [33] aim to model both temporal and structural evolution, but challenges still remain in multi-level propagation [24, 33]. Graph Attention Networks (GATs) and other models [20, 40] have improved predictive performance, with recent research increasingly focused on multi-dimensional aspects, including temporal dynamics, hierarchical structures, and emotional diffusion [12]. Cross-topic propagation models gain attention during the COVID-19 pandemic, examining the timing and sequencing of social media posts [54], but unified frameworks integrating these multi-dimensional features are still lacking [8, 38]. Agent-based simulations, while useful for modeling complex behaviors, face scalability and validation challenges [18, 19, 34]. Despite significant progress, a unified method to simultaneously capture temporal, structural, and multi-dimensional features in opinion propagation remains an open challenge.

## 3 Dataset Construction

To support research on opinion propagation, we collect real-time comments and interaction records from Weibo [50] related to trending events. As illustrated in Figure 2, the dataset exhibits distributions across domains and hierarchical levels. Using specified keywords and event lists, we capture the full opinion cycle, from event inception to attention decline. After data merging, we perform deduplication, standardize formats, and apply a custom dictionary and multilingual detection to remove irrelevant or anomalous

Table 2: Statistics of VISTA Dataset

| Category | Type | Emotion classification |       |       |      |      |      |       |      |       |       |       | Total |        |
|----------|------|------------------------|-------|-------|------|------|------|-------|------|-------|-------|-------|-------|--------|
|          |      | A                      | An    | S     | F    | C    | N    | Ca    | O    | H     | Ex    | El    |       |        |
| Basic    | P    | Normal                 | 2529  | 4479  | 3362 | 1866 | 1552 | 3773  | 429  | 11469 | 9784  | 7343  | 621   | 47207  |
|          |      | Abnormal               | 0     | 0     | 0    | 0    | 0    | 0     | 0    | 0     | 0     | 0     | 0     | 0      |
|          | C    | Normal                 | 11738 | 10477 | 8398 | 9354 | 5409 | 34412 | 4447 | 74039 | 57331 | 37178 | 12587 | 265370 |
|          |      | Abnormal               | 1911  | 1829  | 1006 | 1549 | 1056 | 10873 | 647  | 17443 | 13484 | 9574  | 2273  | 61645  |
|          | R    | Normal                 | 2670  | 2145  | 619  | 1044 | 695  | 10318 | 188  | 2616  | 3341  | 1276  | 45    | 24957  |
|          |      | Abnormal               | 343   | 256   | 49   | 150  | 162  | 2024  | 10   | 564   | 721   | 305   | 37    | 4621   |
| Social   | CL   | Normal                 | 407   | 468   | 311  | 533  | 115  | 1038  | 96   | 2856  | 1265  | 1227  | 161   | 8477   |
|          |      | Abnormal               | 50    | 121   | 60   | 64   | 44   | 236   | 17   | 856   | 495   | 365   | 74    | 2382   |
|          | RL   | Normal                 | 1680  | 1095  | 287  | 641  | 357  | 3730  | 70   | 964   | 1171  | 402   | 16    | 10413  |
|          |      | Abnormal               | 231   | 111   | 25   | 94   | 38   | 710   | 2    | 160   | 241   | 84    | 12    | 1708   |

\*The dataset used in this study does not contain any fake accounts (or astroturfing)

\*P:posts. C:comments. R:replies. CL:comments with likes. RL:replies with likes.

\*A:Angry, An:Anxious, S:Sad, F:Frustrated, C:Consoling, N:Neutral, Ca:Calm, O:Optimistic, H:Happy, Ex:Excited, El:Elated

comments. We also rigorously filter out sensitive personal information. Finally, timestamps are standardized for downstream temporal analysis and hierarchical structure construction.

To retain the content of comments, we reconstruct the tree-like structure of comments and replies by recording and correcting the parent comment IDs. Using hash mapping and indexing, we match each comment with its parent, tracing back layer by layer to the deepest child comment while limiting the depth to a maximum of three levels. This three-level structure captures opinions on posts, comments, and supporting or opposing voices, providing an efficient and comprehensive representation of opinion propagation. In cases where parent IDs are missing or refer to deleted comments, we treat them as top-level comments and mark them separately. After this process, the comment data forms directed trees, with each comment assigned an accurate hierarchical label. As illustrated in Figure 3, the VISTA dataset captures the full structure of opinion events, including replies, comments, and posts. The dataset’s statistical breakdown, categorized by comment levels and emotional annotations, is comprehensively provided in Table 2 for reference.

For sentiment and event label generation, we use the GLM-4-plus model for automated comment annotation, yielding 11 sentiment levels categorized into negative, neutral, and positive, along with the model’s rationale. To ensure annotation reliability, we first manually label a subset of high-quality comments, using them for prompt engineering and fine-tuning before applying batch inference to the remaining comments. During a secondary review, significant discrepancies with manual annotations are corrected. Finally, spot checks are performed to verify dataset quality, achieving a mean Cohen’s Kappa of 0.85 for sentiment annotation across ten final checks. We compare the multidimensional aspects of various opinion propagation datasets in Table 1, highlighting their strengths and limitations. Event labels are also assigned to each comment, along with user attributes, thereby ensuring traceability within the broader social network and public opinion context.

## 4 Task Definition

Suppose there are  $M$  parallel “hot topics” in a given time period, each represented by a root post  $H_m$  for  $m \in \{1, \dots, M\}$ . Under these root posts, users continuously produce hierarchical comments: if a comment replies directly to a hot topic, it is considered level 1;

if it replies to a comment at the previous level, it then immediately spawns a new node in that thread, increasing the level by 1. Formally, we denote each observed comment event as follows:

$$\mathcal{E} = \{e_i \mid e_i = (h_i, l_i, p_i, t_i)\},$$

where  $h_i \in \{1, \dots, M\}$  indicates the hot topic to which the comment belongs ( $i$  represents an individual comment event, which includes its topic, level, parent comment, and timestamp),  $l_i \geq 1$  is the comment’s level within that topic,  $p_i$  is the parent comment or root post, and  $t_i$  is the posting timestamp for that comment.

Because multiple hot topics may exhibit mutual influence (for example, content overlap or shared user bases across different topics), comment evolution can manifest both *cross-topic* mutual excitation and *hierarchical expansion* within each individual chain. Therefore, given a historical comment event set  $\mathcal{E}_{\leq T}$  (i.e., all comments observed up to some time  $T$ ), our goal is to learn a predictive function

$$F(\mathcal{E}_{\leq T}) \mapsto \widehat{\mathcal{E}}_{>T},$$

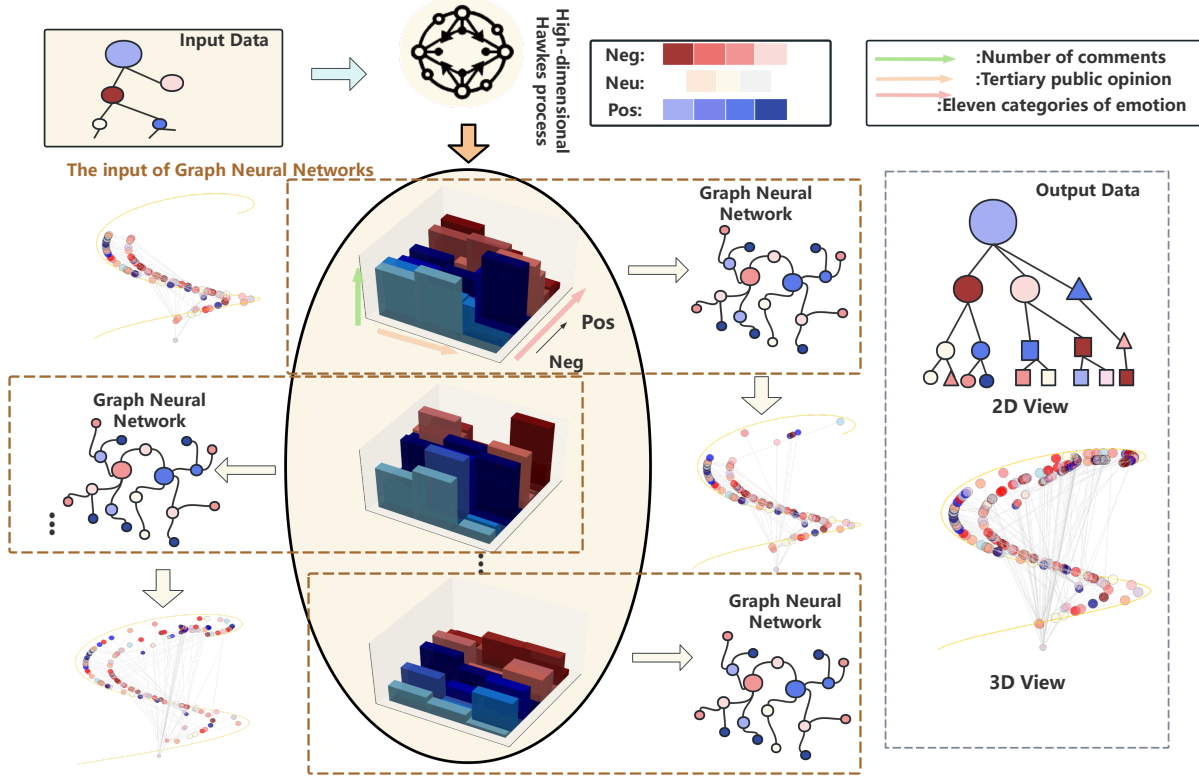
where  $\mathcal{E}_{\leq T}$  denotes, in our formulation, all comments observed at or before time  $T$ , and  $\widehat{\mathcal{E}}_{>T}$  denotes the set of predicted comments that may appear during the subsequent future interval  $(T, T + \Delta]$ .

Concretely, the prediction covers two paired dimensions:

**Temporal dimension**, it focuses on predicting when new comments will emerge and how the volume of comments will change over time across various topics and hierarchical levels.

**Structural dimension**, it focuses on how newly added comments connect to parent nodes, possibly spanning varied topics or levels, thus driving the dynamic evolution of the hierarchical structure.

The core of the task lies in simultaneously accounting for cascading responses within the hierarchical structure (where higher-level comments trigger replies at subsequent levels) and sentiment propagation patterns to provide a comprehensive prediction of future comment behavior. This requires not only modeling the dynamic temporal and structural evolution of the discussion network but also effectively capturing how sentiment propagates through comments, enabling more accurate forecasting of the evolution of public opinion on social media platforms in practice and at scale.



**Figure 4: Architecture of the Social Network Sentiment Analysis Model.** Our framework analyzes sentiment in social networks using a high-dimensional Hawkes process and graph neural networks. The Hawkes process predicts comment volume and sentiment, while GNN learns the social structure, classifies nodes, and constructs a predicted network. Through iterations, the framework achieves sentiment analysis.

## 5 Methodology

As illustrated in Figure 4, the architecture of our Social Network Sentiment Analysis model combines a high-dimensional Hawkes process with a graph neural network to model sentiment dynamics.

Our method integrates multi-level comments with fine-grained sentiment levels. When incorporating the Hawkes process into sentiment modeling, the main challenge lies in obtaining direct predictions for the number of comments associated with different sentiment opinions. To address this, we model the Hawkes process in higher dimensions by combining hierarchy and sentiment levels as two discrete attributes. Each dimension corresponds to a specific “(hierarchy, sentiment level)” pair, enabling the intensity function to estimate the number of comments associated with various sentiment opinions. The high-dimensional Hawkes process captures the temporal evolution of comments under different sentiment labels and hierarchy levels. The output is then used to construct a GNN, further capturing the dynamic propagation structure and sentiment diffusion patterns among comments.

Let  $l$  denote the hierarchy level and  $c$  denote the sentiment level, where there are  $L$  hierarchy levels and  $C$  sentiment levels. In this paper,  $L = 3$  and  $C = 11$ . Each comment is mapped to a dimension  $\omega = (l, c)$ , where  $1 \leq l \leq L$  and  $1 \leq c \leq C$ . The Cartesian product of

hierarchy levels and sentiment levels forms a new set of dimensions:

$$\Omega = \{(l, c) \mid l \in \{1, \dots, L\}, c \in \{1, \dots, C\}\},$$

where  $\Omega$  represents all possible combinations of hierarchy levels and sentiment levels, and  $\omega \in \Omega$  corresponds to a specific type of event. If a comment occurs at time  $t_i^\omega$ , then it is assigned to dimension  $\omega$ . Based on this, the intensity function of the high-dimensional Hawkes process is defined as:

$$\lambda_\omega(t) = \mu_\omega + \sum_{\omega' \in \Omega} \sum_{t_j^{\omega'} \leq t} \alpha_{\omega, \omega'} \phi_{\omega, \omega'}(t - t_j^{\omega'}), \quad (1)$$

where  $\mu_\omega$  is dimension  $\omega$ 's baseline arrival rate,  $\alpha_{\omega, \omega'}$  measures the excitation strength from dimension  $\omega'$  to dimension  $\omega$ , and  $\phi_{\omega, \omega'}(\tau)$  is the decay kernel. Here, the decay kernel function is defined as an exponential kernel,  $\phi_{\omega, \omega'}(\tau) = e^{-\beta_{\omega, \omega'} \tau} \mathbf{1}_{\{\tau \geq 0\}}$ , where  $\beta_{\omega, \omega'}$  is the decay coefficient and  $\mathbf{1}_{\{\tau \geq 0\}}$  is the indicator function.

To estimate  $\mu_\omega$ ,  $\alpha_{\omega, \omega'}$ , and  $\beta_{\omega, \omega'}$ , we employ the maximum likelihood method. Given an observation window  $[0, T]$ , let  $\{t_1^\omega, \dots, t_{N_\omega}^\omega\}$  denote the occurrence times of all events (comments) in dimension  $\omega$ , with  $N_\omega$  being the total number of events. The log-likelihood function of the high-dimensional Hawkes process is expressed as:

$$\mathcal{L} = \sum_{\omega \in \Omega} \left[ \sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega)) - \int_0^T \lambda_\omega(t) dt \right], \quad (2)$$

where  $T$  is the end of the observation window.

### Log-likelihood and Gradient Derivations

The parameters to be learned are:

$$\theta = \{ \{ \mu_\omega \}_{\omega \in \Omega}, \{ \alpha_{\omega, \omega'} \}_{\omega, \omega' \in \Omega}, \{ \beta_{\omega, \omega'} \}_{\omega, \omega' \in \Omega} \}. \quad (3)$$

The gradient of  $\mathcal{L}$  taken with respect to  $\mu_\omega$  is:

$$\frac{\partial \mathcal{L}}{\partial \mu_\omega} = \sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} - \int_0^T 1 dt = \sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} - T. \quad (4)$$

For the gradient with respect to  $\alpha_{\omega, \omega'}$ , it includes log term and integral term. The gradient of the log term can be calculated as:

$$\sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega)) \rightarrow \sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} \cdot \frac{\partial}{\partial \alpha_{\omega, \omega'}} \lambda_\omega(t_i^\omega). \quad (5)$$

For a detailed mathematical proof, refer to Appendix Section B. Using Equation (1), we derive:

$$\frac{\partial}{\partial \alpha_{\omega, \omega'}} \lambda_\omega(t_i^\omega) = \sum_{j=1}^{N_{\omega'}} e^{-\beta_{\omega, \omega'}(t_i^\omega - t_j^{\omega'})} \mathbf{1}_{\{t_j^{\omega'} < t_i^\omega\}}. \quad (6)$$

where  $N_{\omega'}$  is the total number of events of type  $\omega'$  in the observed time window  $[0, T]$ .

The integral term can be rewritten as:

$$\int_0^T \frac{\partial}{\partial \alpha_{\omega, \omega'}} \lambda_\omega(t) dt = \sum_{j=1}^{N_{\omega'}} I_j \quad (7)$$

where  $I_j = \frac{1}{\beta_{\omega, \omega'}} [1 - e^{-\beta_{\omega, \omega'}(T - t_j^{\omega'})}]$ . Combining the terms, we can arrive at the gradient with respect to  $\alpha_{\omega, \omega'}$  as:

$$\frac{\partial \mathcal{L}}{\partial \alpha_{\omega, \omega'}} = \sum_{i=1}^{N_\omega} \frac{\Phi_i}{\lambda_\omega(t_i^\omega)} - \sum_{j=1}^{N_{\omega'}} I_j \quad (8)$$

where  $\Phi_i$  is the excitation sum:  $\Phi_i = \sum_{j=1}^{N_{\omega'}} e^{-\beta_{\omega, \omega'}(t_i^\omega - t_j^{\omega'})} \mathbf{1}_{\{t_j^{\omega'} < t_i^\omega\}}$ ,

and  $I_j$  is the integral term:  $I_j = \frac{1}{\beta_{\omega, \omega'}} [1 - e^{-\beta_{\omega, \omega'}(T - t_j^{\omega'})}]$ .

For a detailed mathematical proof, refer to Appendix Section C. The gradient with respect to  $\beta_{\omega, \omega'}$  follows a similar derivation. For the Hawkes process to remain stable and avoid exponential growth, the following stability condition is required for each dimension  $\omega$ :

$$\sum_{\omega' \in \Omega} \frac{\alpha_{\omega, \omega'}}{\beta_{\omega, \omega'}} < 1. \quad (9)$$

Otherwise, it leads to a critical or supercritical state, resulting in exponential comment growth and loss of stationarity. For a detailed mathematical proof, refer to Appendix Section D.

### Prediction of Comment Counts

To predict the number of comments for a specific hierarchy-sentiment pair within the interval  $[T, T + \Delta]$ , we integrate the intensity function over the interval. For any  $\omega = (l, c)$ , the expected number of events occurring within  $[T, T + \Delta]$  is given by:

$$\hat{N}_\omega(T, T + \Delta) = \int_T^{T+\Delta} \lambda_\omega(t) dt. \quad (10)$$

For scenarios focusing on all sentiment levels within a specific hierarchy level or all hierarchy levels within a specific sentiment level, the predictions for the corresponding dimensions  $\omega$  are aggregated accordingly. The high-dimensional Hawkes model is trained using

stochastic gradient methods, enabling the simultaneous estimation of all parameters  $\mu_\omega$ ,  $\alpha_{\omega, \omega'}$ , and  $\beta_{\omega, \omega'}$ . This framework facilitates both independent and combined calculations of intensities across dimensions during inference or prediction.

### Graph Neural Network Modeling

The GNN models opinion propagation. Its primary tasks are node classification and edge prediction. Node classification predicts each comment's sentiment level, while edge prediction infers propagation paths and their strengths. Using comments and sentiment distributions predicted by the high-dimensional Hawkes process, we construct an opinion-propagation graph, denoted as  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , where  $\mathcal{V}$  represents the set of comment nodes, and  $\mathcal{E}$  represents the parent-child relationships between comments.

Each node's features are composed of the intensity function values  $\lambda_\omega(t)$  generated by the Hawkes process and the sentiment distribution  $q_c(v)$ . In this formulation,  $\lambda_\omega(t)$  represents the arrival intensity of the comment, and  $q_c(v)$  denotes the probability that node  $v$  belongs to sentiment level  $c \in \{1, 2, \dots, C\}$ . Specifically, the sentiment distribution is formally defined as follows:

$$q_c(v) = \frac{\lambda_{(l,c)}(t)}{\sum_{c'=1}^C \lambda_{(l,c')}(t)}, \quad (11)$$

where  $\lambda_{(l,c)}(t)$  is the intensity function value at time  $t$  for hierarchy level  $l$  and sentiment level  $c$ , and  $C$  is the total number of sentiment categories. This distribution captures the sentiment inclination of the node. The edge features  $\mathbf{e}_{uv}$  include the time difference between nodes,  $\Delta t = t_v - t_u$ ,  $\Delta t > 0$ , and the excitation strength  $\alpha_{\omega, \omega'}$ . The former measures the time interval between comments, while the latter reflects the propagation relationships between different sentiments and hierarchy levels more precisely.

Within the constructed propagation graph, node representations are updated through an iterative message-passing scheme. Let  $t$  denote the current time slice and the embedding of node  $v$  be defined as  $\mathbf{h}_v^{(t)}$ . The updated rule for node embeddings is:

$$\mathbf{h}_v^{(t+1)} = \sigma \left( \mathbf{W}_1 \mathbf{h}_v^{(t)} + \sum_{u \in \mathcal{N}(v)} \mathbf{W}_2 \mathbf{e}_{uv} \mathbf{h}_u^{(t)} \right), \quad (12)$$

where  $\mathbf{h}_v^{(t+1)}$  is the embedding of node  $v$  at time  $t + 1$ ,  $\mathcal{N}(v)$  is the set of neighbors of node  $v$ ,  $\mathbf{e}_{uv}$  is the feature vector of edge  $(u, v)$ ,  $\mathbf{W}_1, \mathbf{W}_2$  are learnable linear transformation matrices, and  $\sigma(\cdot)$  is a non-linear activation function. This update aggregates features by combining the node's historical embedding with neighbor embeddings, appropriately weighted by edge features.

Based on the updated node embeddings, we classify the sentiment level of each node using the following formula:

$$P(c | v) = \text{softmax}(\mathbf{W}_c \mathbf{h}_v^{(t+1)}), \quad (13)$$

where  $\mathbf{W}_c$  is the linear transformation matrix for sentiment classification, and  $P(c | v)$  represents the predicted probability that node  $v$  belongs to sentiment level  $c$  in our model.

To optimize the accuracy of node classification and edge prediction, we introduce two loss functions: the sentiment prediction loss  $\mathcal{L}_{\text{sentiment}}$  and the graph structure loss  $\mathcal{L}_{\text{struct}}$ . The sentiment prediction loss measures the accuracy of the predicted sentiment

**Table 3: Baseline Methods Comparison on VISTA Dataset**

| Method                     | Sentiment Accuracy (Val/Test) |                  |                  |                  | Structural Consistency Accuracy (Val/Test) |                  |                  |                  |
|----------------------------|-------------------------------|------------------|------------------|------------------|--------------------------------------------|------------------|------------------|------------------|
|                            | 15%                           | 20%              | 25%              | 30%              | 15%                                        | 20%              | 25%              | 30%              |
| CNN [28]                   | 18.7/17.3                     | 24.5/22.8        | 28.1/26.2        | 29.8/27.6        | -/-                                        | -/-              | -/-              | -/-              |
| LSTM [21]                  | 20.5/19.5                     | 27.6/25.1        | 31.4/29.1        | 33.2/30.8        | -/-                                        | -/-              | -/-              | -/-              |
| GRU [13]                   | 21.6/20.3                     | 28.4/26.6        | 32.1/29.8        | 33.9/31.4        | -/-                                        | -/-              | -/-              | -/-              |
| Transformer [46]           | 24.4/22.9                     | 31.2/29.1        | 34.8/32.5        | 36.1/33.7        | -/-                                        | -/-              | -/-              | -/-              |
| GCN [26]                   | 19.3/18.0                     | 25.2/23.4        | 28.7/26.8        | 30.1/28.0        | -/-                                        | -/-              | -/-              | -/-              |
| GAT [47]                   | 22.3/21.1                     | 29.4/27.2        | 33.1/30.9        | 34.6/32.3        | -/-                                        | -/-              | -/-              | -/-              |
| GraphSAINT [55]            | 21.3/20.0                     | 27.9/26.1        | 31.6/29.4        | 33.0/30.7        | -/-                                        | -/-              | -/-              | -/-              |
| TGN [41]                   | 26.7/24.9                     | 34.3/32.0        | 37.9/35.6        | 39.2/36.8        | 16.1/14.9                                  | 22.4/20.6        | 26.1/24.3        | 27.8/25.9        |
| DyRep [44]                 | 26.1/24.3                     | 33.2/31.2        | 36.8/34.7        | 38.1/35.9        | 15.6/14.5                                  | 21.7/19.8        | 25.3/23.6        | 26.9/25.1        |
| EvolveGCN [37]             | 24.6/22.8                     | 31.7/29.7        | 35.2/33.1        | 36.4/34.2        | 14.2/13.1                                  | 19.8/18.2        | 23.1/21.7        | 24.6/23.0        |
| Multi-state IC [25]        | 14.7/13.5                     | 20.5/19.1        | 23.6/22.1        | 24.9/23.2        | 8.2/7.6                                    | 12.4/11.5        | 15.1/14.2        | 16.3/15.2        |
| <b>Hawkes-Graph (Ours)</b> | <b>32.7/30.2</b>              | <b>39.8/36.7</b> | <b>43.2/40.1</b> | <b>44.9/41.6</b> | <b>18.1/16.8</b>                           | <b>24.7/22.9</b> | <b>28.3/26.5</b> | <b>30.1/28.2</b> |

\* This table presents the evaluation of the model’s performance on future opinion chain prediction tasks, based on three different data proportions: 15%, 20%, 25% and 30%.

**Table 4: Ablation Study on Hawkes Process Parameter**

| Parameter Configuration                            | SA (25%)         | SCA (25%)        |
|----------------------------------------------------|------------------|------------------|
| Full Interaction Matrix (Ours)                     | <b>43.2/40.1</b> | <b>28.3/26.5</b> |
| Diagonal Only                                      | 38.1/35.7        | 22.9/21.4        |
| Block Diagonal                                     | 40.3/37.8        | 25.1/23.6        |
| Shared Decay ( $\beta_{\omega, \omega'} = \beta$ ) | 41.7/38.9        | 26.8/25.1        |
| No Cross-Emotion Excitation                        | 39.5/37.2        | 24.6/23.1        |
| Uniform Baseline ( $\mu_{\omega} = \mu$ )          | 42.1/39.4        | 27.1/25.4        |

**Table 5: Ablation Study on Temporal Kernel Functions**

| Kernel Function    | SA (25%)         | SCA (25%)        |
|--------------------|------------------|------------------|
| Exponential (Ours) | <b>43.2/40.1</b> | <b>28.3/26.5</b> |
| Constant           | 35.7/33.4        | 21.2/19.8        |
| Step Function      | 39.3/37.1        | 24.5/23.0        |
| Weibull            | 42.1/39.7        | 27.1/25.4        |

distribution for nodes and is formally defined as:

$$\mathcal{L}_{\text{sentiment}} = - \sum_{v \in \mathcal{V}_{\text{pred}}} \sum_{c=1}^C q_c(v) \cdot \ln P(c | v), \quad (14)$$

where  $\mathcal{V}_{\text{pred}}$  denotes the set of nodes predicted by the Hawkes process, and  $q_c(v)$  is the normalized sentiment distribution from the Hawkes process, serving as the true sentiment label of node  $v$ .

Graph-structure loss quantifies differences between the predicted graph  $\mathcal{G}_{\text{pred}}$  and the true graph  $\mathcal{G}_{\text{true}}$ . It is defined as:

$$\mathcal{L}_{\text{struct}} = \sum_{(u,v) \in \mathcal{E}} |e_{uv}^{\text{pred}} - e_{uv}^{\text{true}}|, \quad (15)$$

where  $\mathcal{E}$  is the set of edges in the graph, and  $e_{uv}^{\text{pred}}$  and  $e_{uv}^{\text{true}}$  represent the feature values of edge  $(u, v)$  in the predicted graph and the true graph, respectively, reflecting the differences in propagation paths.

The final optimization objective combines the two losses, and the composite loss function is defined as:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{sentiment}} + \lambda_2 \mathcal{L}_{\text{struct}}, \quad (16)$$

where  $\lambda_1$  and  $\lambda_2$  are hyperparameters that control the weights of the sentiment loss and the structure loss, respectively. By minimizing  $\mathcal{L}_{\text{total}}$ , the graph neural network more effectively optimizes the performance of both node classification and edge prediction, thereby capturing the dynamic evolution of opinion propagation in terms of both temporal and structural dimensions.

## 6 Experiments

### 6.1 Experiments Setup

We partition VISTA into 127 training, 16 validation, and 16 test topics. Temporal prediction uses 15%, 20%, 25%, and 30% historical event ratios to forecast future propagation dynamics. The high-dimensional Hawkes process operates on 33 dimensions (3 hierarchical levels  $\times$  11 emotions) with exponential decay kernels. Training proceeds for 30 epochs using SGD (lr=0.001) with validation every 5 epochs. The GNN employs 3-layer graph convolution (hidden dim=128, batch=8, dropout=0.2). Implementation uses PyTorch on NVIDIA V100. Parameters initialize as:  $\mu_{\omega}, \alpha_{\omega, \omega'} \sim \mathcal{N}(0, 0.01)$ ,  $\beta_{\omega, \omega'} \sim \mathcal{U}[0.05, 0.15]$ .

### 6.2 Baseline for Dynamic Propagation

Table 3 establishes comprehensive baseline performance for the novel multi-dimensional opinion propagation prediction task on VISTA. We evaluate 11 representative methods spanning traditional deep learning (CNN, LSTM, Transformer), static graph networks (GCN, GAT, GraphSAINT), temporal graph methods (TGN, DyRep, EvolveGCN), and propagation models (Multi-state IC). Across 15%-30% historical data ratios, our Hawkes-Graph framework significantly outperforms all baselines on sentiment accuracy (SA), achieving up to 44.9%/41.6% and improving 14.5%/12.9% over the best baseline TGN. For structural consistency accuracy (SCA), traditional methods cannot handle dynamic structure prediction, while our approach achieves 30.1%/28.2%, establishing effective benchmarks for this new task.

**Table 6: Parameter Pattern Discovery in High-dimensional Hawkes Process**

| Pattern Analysis               | Source $\omega$ | Target $\omega'$ | $\alpha_{\omega,\omega'}$ | $\beta_{\omega,\omega'}$ | $\mu_{\omega'}$ | Half-life (h) | Excitation Ratio | Pattern Type               |
|--------------------------------|-----------------|------------------|---------------------------|--------------------------|-----------------|---------------|------------------|----------------------------|
| <b>Emotional Propagation</b>   | (L1, Angry)     | (L1, Angry)      | 0.847                     | 0.089                    | 0.124           | 7.78          | 9.52             | Self-reinforcement         |
|                                | (L1, Happy)     | (L1, Happy)      | 0.792                     | 0.156                    | 0.098           | 4.44          | 5.08             | Self-reinforcement         |
|                                | (L1, Sad)       | (L1, Sad)        | 0.718                     | 0.203                    | 0.089           | 3.41          | 3.54             | Self-reinforcement         |
|                                | (L1, Happy)     | (L1, Excited)    | 0.634                     | 0.134                    | 0.145           | 5.17          | 4.73             | Cross-emotional excitation |
|                                | (L1, Angry)     | (L1, Sad)        | 0.389                     | 0.178                    | 0.089           | 3.89          | 2.19             | Negative emotion cascade   |
|                                | (L1, Sad)       | (L1, Angry)      | 0.312                     | 0.167                    | 0.124           | 4.15          | 1.87             | Bidirectional negativity   |
|                                | (L1, Angry)     | (L1, Happy)      | 0.124                     | 0.245                    | 0.098           | 2.83          | 0.51             | Emotional opposition       |
|                                | (L1, Excited)   | (L1, Happy)      | 0.587                     | 0.189                    | 0.098           | 3.67          | 3.11             | Positive emotion loop      |
|                                | (L1, Neutral)   | (L1, Angry)      | 0.203                     | 0.198                    | 0.124           | 3.50          | 1.03             | Neutral-to-negative        |
|                                | (L1, Neutral)   | (L1, Happy)      | 0.298                     | 0.187                    | 0.098           | 3.71          | 1.59             | Neutral-to-positive        |
| <b>Hierarchical Excitation</b> | (L1, Angry)     | (L2, Angry)      | 0.782                     | 0.134                    | 0.089           | 5.17          | 5.84             | Downward propagation       |
|                                | (L2, Angry)     | (L3, Angry)      | 0.689                     | 0.156                    | 0.067           | 4.44          | 4.42             | Continued propagation      |
|                                | (L1, Happy)     | (L3, Happy)      | 0.456                     | 0.189                    | 0.078           | 3.67          | 2.41             | Skip-level propagation     |
|                                | (L2, Angry)     | (L1, Angry)      | 0.334                     | 0.198                    | 0.124           | 3.50          | 1.69             | Upward feedback            |
|                                | (L3, Happy)     | (L2, Happy)      | 0.372                     | 0.234                    | 0.098           | 2.96          | 1.59             | Level-reversal feedback    |
|                                | (L3, Sad)       | (L1, Sad)        | 0.273                     | 0.267                    | 0.089           | 2.60          | 1.02             | Cross-level feedback       |

$\omega = (l, c)$  represents a combination of hierarchy level  $l$  and emotion category  $c$ . Half-life indicates the time for excitation effects to decay to 50% ( $\ln(2)/\beta_{\omega,\omega'}$ ), while Excitation Ratio measures the relative strength of excitation to decay ( $\alpha_{\omega,\omega'}/\beta_{\omega,\omega'}$ ).

**Table 7: Statistical Validation of Mechanistic Patterns.**

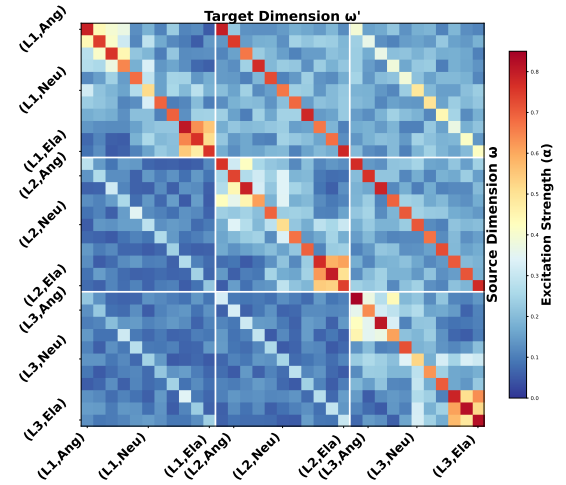
| Mechanism & Test                                   | Statistic             | p-value     | Effect Size     |
|----------------------------------------------------|-----------------------|-------------|-----------------|
| <b>Differential Emotional Propagation Strength</b> |                       |             |                 |
| Angry vs. Happy                                    | $t = 12.34$           | $p < 0.001$ | $d = 0.84$      |
| Happy vs. Sad                                      | $t = 8.67$            | $p < 0.001$ | $d = 0.72$      |
| Positive vs. Negative                              | $t = 6.92$            | $p < 0.01$  | $d = 0.58$      |
| Overall ANOVA                                      | $F(10, 1589) = 15.67$ | $p < 0.001$ | $\eta^2 = 0.28$ |
| <b>Asymmetric Hierarchical Excitation</b>          |                       |             |                 |
| L1→L2 vs. L2→L1                                    | $t = 9.32$            | $p < 0.001$ | $d = 1.24$      |
| L2→L3 vs. L3→L2                                    | $t = 5.74$            | $p < 0.01$  | $d = 0.87$      |
| L1→L3 vs. L3→L1                                    | $t = 3.43$            | $p < 0.05$  | $d = 0.65$      |
| Wilcoxon Rank Test                                 | $Z = 4.89$            | $p < 0.01$  | $r = 0.58$      |

Tests validate differential emotional propagation strength and asymmetric hierarchical excitation using ANOVA and paired t-tests.

### 6.3 Ablation Study

We first evaluate the structural components of our model. As shown in Table 4, experiments on Hawkes process parameter modeling demonstrate the necessity of full interaction matrices. Simplified diagonal or block-diagonal structures sever cross-dimensional dependencies, significantly degrading performance; notably, the removal of cross-emotion excitation confirms that emotional cascades are the core driver of opinion dynamics. Furthermore, Table 5 indicates that exponential decay kernels outperform constant, step, and Weibull functions, offering the most mathematically elegant and accurate capture of memory decay in opinion propagation.

Tables 6 and 7 jointly quantify the intrinsic mechanisms of opinion dynamics through parameter analysis and statistical testing. First, we observe *differential emotional propagation*: negative emotions exhibit strong self-reinforcement (excitation ratio of 9.52), whereas positive emotions favor cross-category propagation. This disparity is statistically significant (angry vs. happy:  $t = 12.34, p < 0.001, d = 0.84$ ), with ANOVA results ( $F(10, 1589) = 15.67, p < 0.001$ ) confirming the dominant self-excitation of negative sentiments. Second, *hierarchical asymmetry* is evident: downward propagation (L1→L2 ratio 5.84) significantly outweighs upward feedback (ratio 1.69). This structural bias is confirmed by statistical tests ( $t = 9.32, p < 0.001, d = 1.24$ ), indicating root-level content exerts

**Figure 5: Excitation strength matrix visualization showing propagation patterns across 33 dimensions.**

stronger influence on deeper replies. Third, *temporal memory effects* vary by interaction type, where emotional self-reinforcement maintains the longest memory span compared to the faster decay of cross-dimensional interactions.

Figure 5 visualizes the  $33 \times 33$  excitation strength matrix, corroborating the above findings. The heatmap reveals: (1) pronounced diagonal self-reinforcement, particularly for negative emotions; (2) block-structured off-diagonal regions showing gradient decay from L1 to L3; and (3) the role of neutral emotion as a bridging hub with uniform excitation. Visualizations reveal positive emotions spread mutually, while negative emotions self-persist.

## 7 CONCLUSION

We propose an interpretable modeling paradigm using the VISTA dataset and Hawkes-Graph framework, addressing mechanistic limitations of existing methods. We reveal three core mechanisms differential emotional propagation, asymmetric hierarchical excitation, and temporal memory—enabling transition from prediction to mechanistic explanation of multi-dimensional opinion dynamics.

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## Appendix

### A Evaluation Metrics

**Sentiment Prediction Accuracy (SA):** The sentiment prediction accuracy measures how well the model predicts the sentiment label for each comment. Given the true sentiment label  $c(v)$  and the predicted sentiment label  $\hat{c}(v)$  for each comment node  $v$ , the sentiment prediction accuracy  $\text{Acc}_{\text{sentiment}}$  is defined as:

$$\text{Acc}_{\text{sentiment}} = \frac{1}{N} \sum_{v \in \mathcal{V}} \mathbf{1}(\hat{c}(v) = c(v))$$

where  $\mathbf{1}(\cdot)$  is an indicator function that equals 1 when the predicted sentiment matches the true sentiment, and 0 otherwise.

**Structural Consistency Prediction Accuracy (SCA):** The structural consistency prediction accuracy measures how well the model predicts the hierarchical structure of comments. Specifically, it checks whether, in our evaluation, the set of child nodes for each parent node is correctly predicted. The structural consistency prediction accuracy  $\text{Acc}_{\text{structure}}$  is defined as:

$$\text{Acc}_{\text{structure}} = \frac{1}{|\mathcal{V}_{\text{true}}|} \sum_{v \in \mathcal{V}_{\text{true}}} \mathbf{1}(C_v^{\text{pred}} = C_v^{\text{true}})$$

where  $C_v^{\text{true}}$  and  $C_v^{\text{pred}}$  are the sets of child nodes of the parent node  $v$  in the true and predicted graphs, respectively.

### B Gradient Derivation for Parameter $\mu_\omega$

Given the high-dimensional Hawkes process intensity function  $\lambda_\omega(t)$ , the log-likelihood function is defined as:

$$\mathcal{L} = \sum_{\omega \in \Omega} \left[ \sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega)) - \int_0^T \lambda_\omega(t) dt \right], \quad (17)$$

where  $t_i^\omega$  is the  $i$ -th event time in the  $\omega$ -th dimension.

The intensity function  $\lambda_\omega(t)$  is composed of two terms: the first term  $\sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega))$  is the log-likelihood of the observed event times  $t_i^\omega$ , and the second term  $\int_0^T \lambda_\omega(t) dt$  is the integral of the intensity function  $\lambda_\omega(t)$  over the time interval  $[0, T]$ , representing the expected number of events.

The intensity function for the high-dimensional Hawkes process is defined as:

$$\lambda_\omega(t) = \mu_\omega + \sum_{\omega' \in \Omega} \sum_{t_j^{\omega'} \leq t} \alpha_{\omega, \omega'} \phi_{\omega, \omega'}(t - t_j^{\omega'}), \quad (18)$$

where  $\mu_\omega$  is the baseline intensity, representing the event rate in the absence of any historical events,  $\alpha_{\omega, \omega'}$  is the excitation strength from event type  $\omega'$  to  $\omega$ , and  $\phi_{\omega, \omega'}(t - t_j^{\omega'})$  is the decay kernel, typically in the form of an exponential decay.

We need to compute the gradient of the log-likelihood function with respect to  $\mu_\omega$ :

$$\frac{\partial \mathcal{L}}{\partial \mu_\omega}. \quad (19)$$

The gradient with respect to  $\mu_\omega$  is derived in two parts. First, we consider the left part:

$$\sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega)). \quad (20)$$

By taking the partial derivative with respect to  $\mu_\omega$  using the chain rule, we obtain that:

$$\frac{\partial}{\partial \mu_\omega} \ln(\lambda_\omega(t_i^\omega)) = \frac{1}{\lambda_\omega(t_i^\omega)} \cdot \frac{\partial \lambda_\omega(t_i^\omega)}{\partial \mu_\omega}. \quad (21)$$

Since  $\lambda_\omega(t)$  only involves  $\mu_\omega$  in the first term (i.e.,  $\mu_\omega$  only affects the baseline part of the intensity function), it yields that

$$\frac{\partial \lambda_\omega(t)}{\partial \mu_\omega} = 1. \quad (22)$$

Therefore, for each event  $t_i^\omega$ , we conclude:

$$\frac{\partial}{\partial \mu_\omega} \ln(\lambda_\omega(t_i^\omega)) = \frac{1}{\lambda_\omega(t_i^\omega)}. \quad (23)$$

Summing over all  $i$ , we get:

$$\sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)}. \quad (24)$$

Next, we will tackle the second part:

$$- \int_0^T \lambda_\omega(t) dt. \quad (25)$$

We note that the above term involves the integral of  $\lambda_\omega(t)$  which is a function of  $\mu_\omega$ . Taking the partial derivative with respect to  $\mu_\omega$ , we get:

$$\frac{\partial}{\partial \mu_\omega} \left( - \int_0^T \lambda_\omega(t) dt \right) = - \int_0^T \frac{\partial \lambda_\omega(t)}{\partial \mu_\omega} dt. \quad (26)$$

By the simple calculation,  $\frac{\partial \lambda_\omega(t)}{\partial \mu_\omega} = 1$ , we arrive at:

$$- \int_0^T 1 dt = -T. \quad (27)$$

Combining the two parts, we obtain the gradient:

$$\frac{\partial \mathcal{L}}{\partial \mu_\omega} = \sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} - T. \quad (28)$$

Thus, the gradient of the log-likelihood function  $\mathcal{L}$  with respect to  $\mu_\omega$  is given by:

$$\frac{\partial \mathcal{L}}{\partial \mu_\omega} = \sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} - T, \quad (29)$$

which can be interpreted as follows: the first term  $\sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)}$  is the gradient based on the actual events, reflecting the relationship

between the observed event times and the intensity function; the second term  $-T$  is the penalty term, which reflects the contribution of the expected number of events over the entire time window, ensuring that the model does not predict excessively high event frequencies during periods without events.

### C Gradient Derivation for Parameter $\alpha_{\omega, \omega'}$

The log-likelihood function for the high-dimensional Hawkes process involves two main terms: the log term and the integral term. The log-likelihood function for the parameter  $\alpha_{\omega, \omega'}$  is given by:

$$\mathcal{L} = \sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega)) - \int_0^T \lambda_\omega(t) dt, \quad (30)$$

where  $\lambda_\omega(t)$  is the intensity function for the process.

#### Step 1: Log Term Gradient

Recall that the log term is defined as:

$$\sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega)). \quad (31)$$

Taking the derivative with respect to  $\alpha_{\omega, \omega'}$ , we deduce that:

$$\frac{\partial}{\partial \alpha_{\omega, \omega'}} \ln(\lambda_\omega(t_i^\omega)) = \frac{1}{\lambda_\omega(t_i^\omega)} \cdot \frac{\partial \lambda_\omega(t_i^\omega)}{\partial \alpha_{\omega, \omega'}}. \quad (32)$$

We then substitute the intensity function into the above equation:

$$\lambda_\omega(t) = \mu_\omega + \sum_{\omega' \in \Omega} \sum_{t_j^{\omega'} \leq t} \alpha_{\omega, \omega'} \phi_{\omega, \omega'}(t - t_j^{\omega'}), \quad (33)$$

and calculate the derivative of  $\lambda_\omega(t)$  with respect to  $\alpha_{\omega, \omega'}$  as:

$$\frac{\partial \lambda_\omega(t)}{\partial \alpha_{\omega, \omega'}} = \sum_{t_j^{\omega'} \leq t} \phi_{\omega, \omega'}(t - t_j^{\omega'}). \quad (34)$$

Thus, the gradient of the log term is:

$$\frac{\partial}{\partial \alpha_{\omega, \omega'}} \ln(\lambda_\omega(t_i^\omega)) = \frac{1}{\lambda_\omega(t_i^\omega)} \cdot \sum_{j=1}^{N_{\omega'}} e^{-\beta_{\omega, \omega'}(t_i^\omega - t_j^{\omega'})} \mathbf{1}_{\{t_j^{\omega'} < t_i^\omega\}}. \quad (35)$$

The full gradient for the log term is:

$$\sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} \sum_{j=1}^{N_{\omega'}} e^{-\beta_{\omega, \omega'}(t_i^\omega - t_j^{\omega'})} \mathbf{1}_{\{t_j^{\omega'} < t_i^\omega\}}. \quad (36)$$

#### Step 2: Integral Term Gradient

Recall that the integral term is given by:

$$- \int_0^T \lambda_\omega(t) dt. \quad (37)$$

The derivative with respect to  $\alpha_{\omega, \omega'}$  is:

$$\frac{\partial}{\partial \alpha_{\omega, \omega'}} \left( - \int_0^T \lambda_\omega(t) dt \right) = - \int_0^T \frac{\partial \lambda_\omega(t)}{\partial \alpha_{\omega, \omega'}} dt. \quad (38)$$

We calculate the derivative of  $\lambda_\omega(t)$  with respect to  $\alpha_{\omega, \omega'}$  and get:

$$\frac{\partial \lambda_\omega(t)}{\partial \alpha_{\omega, \omega'}} = \sum_{t_j^{\omega'} \leq t} e^{-\beta_{\omega, \omega'}(t - t_j^{\omega'})} \mathbf{1}_{\{t \geq t_j^{\omega'}\}}. \quad (39)$$

Therefore, the gradient of the integral term can be written as:

$$- \int_0^T \sum_{t_j^{\omega'} \leq t} e^{-\beta_{\omega, \omega'}(t - t_j^{\omega'})} \mathbf{1}_{\{t \geq t_j^{\omega'}\}} dt. \quad (40)$$

We then replace the indicator function and perform the integral from  $t_j^{\omega'}$  to  $T$ :

$$\int_{t_j^{\omega'}}^T e^{-\beta_{\omega, \omega'}(t - t_j^{\omega'})} dt = \frac{1}{\beta_{\omega, \omega'}} \left( 1 - e^{-\beta_{\omega, \omega'}(T - t_j^{\omega'})} \right). \quad (41)$$

Thus, the gradient of the integral term is calculated as:

$$- \sum_{j=1}^{N_{\omega'}} \frac{1}{\beta_{\omega, \omega'}} \left[ 1 - e^{-\beta_{\omega, \omega'}(T - t_j^{\omega'})} \right]. \quad (42)$$

#### Step 3: Final Gradient Expression

Combining the log term and integral term gradients, we obtain the complete gradient with respect to  $\alpha_{\omega, \omega'}$  as:

$$\frac{\partial \mathcal{L}}{\partial \alpha_{\omega, \omega'}} = \sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} \sum_{j=1}^{N_{\omega'}} e^{-\beta_{\omega, \omega'}(t_i^\omega - t_j^{\omega'})} \mathbf{1}_{\{t_j^{\omega'} < t_i^\omega\}} - \quad (43)$$

$$\sum_{j=1}^{N_{\omega'}} \frac{1}{\beta_{\omega, \omega'}} \left[ 1 - e^{-\beta_{\omega, \omega'}(T - t_j^{\omega'})} \right]. \quad (44)$$

### D Gradient Derivation for Parameter $\beta_{\omega, \omega'}$

The log-likelihood function for the high-dimensional Hawkes process involves two main terms: the log term and the integral term. The log-likelihood function for the parameter  $\alpha_{\omega, \omega'}$  is given by:

$$\mathcal{L} = \sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega)) - \int_0^T \lambda_\omega(t) dt, \quad (45)$$

where  $\lambda_\omega(t)$  is the intensity function for the process.

#### Step 1: Log Term Gradient

The log term is:

$$\sum_{i=1}^{N_\omega} \ln(\lambda_\omega(t_i^\omega)). \quad (46)$$

Taking the derivative with respect to  $\alpha_{\omega, \omega'}$ :

$$\frac{\partial}{\partial \alpha_{\omega, \omega'}} \ln(\lambda_\omega(t_i^\omega)) = \frac{1}{\lambda_\omega(t_i^\omega)} \cdot \frac{\partial \lambda_\omega(t_i^\omega)}{\partial \alpha_{\omega, \omega'}}. \quad (47)$$

Using the intensity function:

$$\lambda_\omega(t) = \mu_\omega + \sum_{\omega' \in \Omega} \sum_{t_j^{\omega'} \leq t} \alpha_{\omega, \omega'} \phi_{\omega, \omega'}(t - t_j^{\omega'}). \quad (48)$$

The derivative of  $\lambda_\omega(t)$  with respect to  $\alpha_{\omega, \omega'}$  is:

$$\frac{\partial \lambda_\omega(t)}{\partial \alpha_{\omega, \omega'}} = \sum_{t_j^{\omega'} \leq t} \phi_{\omega, \omega'}(t - t_j^{\omega'}). \quad (49)$$

Thus, the gradient of the log term is:

$$\frac{\partial}{\partial \alpha_{\omega, \omega'}} \ln(\lambda_\omega(t_i^\omega)) = \frac{1}{\lambda_\omega(t_i^\omega)} \cdot \sum_{j=1}^{N_{\omega'}} e^{-\beta_{\omega, \omega'}(t_i^\omega - t_j^{\omega'})} \mathbf{1}_{\{t_j^{\omega'} < t_i^\omega\}}. \quad (50)$$

The full gradient for the log term is:

$$\sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} \sum_{j=1}^{N_{\omega'}} e^{-\beta_{\omega, \omega'}(t_i^\omega - t_j^{\omega'})} \mathbf{1}_{\{t_j^{\omega'} < t_i^\omega\}}. \quad (51)$$

## Step 2: Integral Term Gradient

The integral term is:

$$- \int_0^T \lambda_\omega(t) dt. \quad (52)$$

The derivative with respect to  $\alpha_{\omega, \omega'}$  is:

$$\frac{\partial}{\partial \alpha_{\omega, \omega'}} \left( - \int_0^T \lambda_\omega(t) dt \right) = - \int_0^T \frac{\partial \lambda_\omega(t)}{\partial \alpha_{\omega, \omega'}} dt. \quad (53)$$

We know that:

$$\frac{\partial \lambda_\omega(t)}{\partial \alpha_{\omega, \omega'}} = \sum_{t_j^{\omega'} \leq t} e^{-\beta_{\omega, \omega'}(t - t_j^{\omega'})} \mathbf{1}_{\{t \geq t_j^{\omega'}\}}. \quad (54)$$

Therefore, the gradient of the integral term is:

$$- \int_0^T \sum_{t_j^{\omega'} \leq t} e^{-\beta_{\omega, \omega'}(t - t_j^{\omega'})} \mathbf{1}_{\{t \geq t_j^{\omega'}\}} dt. \quad (55)$$

We perform the integral from  $t_j^{\omega'}$  to  $T$ :

$$\int_{t_j^{\omega'}}^T e^{-\beta_{\omega, \omega'}(t - t_j^{\omega'})} dt = \frac{1}{\beta_{\omega, \omega'}} \left( 1 - e^{-\beta_{\omega, \omega'}(T - t_j^{\omega'})} \right). \quad (56)$$

Thus, the gradient of the integral term is:

$$- \sum_{j=1}^{N_{\omega'}} \frac{1}{\beta_{\omega, \omega'}} \left[ 1 - e^{-\beta_{\omega, \omega'}(T - t_j^{\omega'})} \right]. \quad (57)$$

## Final Gradient Expression

Combining the log term and integral term gradients, the complete gradient with respect to  $\alpha_{\omega, \omega'}$  is:

$$\frac{\partial \mathcal{L}}{\partial \alpha_{\omega, \omega'}} = \sum_{i=1}^{N_\omega} \frac{1}{\lambda_\omega(t_i^\omega)} \sum_{j=1}^{N_{\omega'}} e^{-\beta_{\omega, \omega'}(t_i^\omega - t_j^{\omega'})} \mathbf{1}_{\{t_j^{\omega'} < t_i^\omega\}} - \quad (58)$$

$$\sum_{j=1}^{N_{\omega'}} \frac{1}{\beta_{\omega, \omega'}} \left[ 1 - e^{-\beta_{\omega, \omega'}(T - t_j^{\omega'})} \right]. \quad (59)$$